

An Analysis of Linear regression

Linear regression

$$\hat{y} = X\beta + \varepsilon$$

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

Linear regression -- Part 1

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_p X_{ip} + \varepsilon_i$$

Linear regression -- Part 2

for all observations indexed as $i = 1, \dots, n$ and for all dependent variables indexed as $j = 1, \dots, m$. Nearly all real-world regression models involve multiple predictors, and basic descriptions of linear regression are often phrased in terms of the multiple regression model. Note, however, that in these cases the response variable y is still a scalar. Another term, multivariate linear regression, refers to cases where y is a vector, i.e., the same as general linear regression.

Linear regression -- Part 3

Now, we need to look for a parameter that maximizes this likelihood function. Since the logarithmic function is strictly increasing, instead of maximizing this function, we can also maximize its logarithm and find the optimal parameter that way.

Linear regression -- Part 4

$$\begin{aligned} \arg \max_{\beta} \rightarrow I(D, \beta) &= \arg \max_{\beta} \rightarrow (n \log \frac{1}{2\pi\sigma^2} \sum_{i=1}^n (y_i - \beta^T x_i)^2) = \arg \min_{\beta} \rightarrow \sum_{i=1}^n (y_i - \beta^T x_i)^2 = \arg \min_{\beta} \rightarrow L(D, \beta) \\ \arg \max_{\beta} \rightarrow I(D, \beta) &= \arg \min_{\beta} \rightarrow \sum_{i=1}^n (y_i - \beta^T x_i)^2 \\ \arg \min_{\beta} \rightarrow L(D, \beta) &= \arg \min_{\beta} \rightarrow \sum_{i=1}^n (y_i - \beta^T x_i)^2 \end{aligned}$$

Linear regression -- Part 5

$$\xi^T(w) = w_1 \beta_1 + w_2 \beta_2 + \dots + w_q \beta_q$$

Linear regression -- Part 6

Building science Linear regression is commonly used in building science field studies to derive characteristics of building occupants. In a thermal comfort field study, building scientists usually ask occupants' thermal sensation votes, which range from -3 (feeling cold) to 0 (neutral) to +3 (feeling hot), and measure occupants' surrounding temperature data. A neutral or comfort temperature can be calculated based on a linear regression between the thermal sensation vote and indoor temperature, and setting the thermal sensation vote as zero. However, there has been a debate on the regression direction: regressing thermal sensation votes (y-axis) against indoor temperature (x-axis) or the opposite: regressing indoor temperature (y-axis) against thermal sensation votes (x-axis).

Linear regression -- Part 7

Estimation methods A large number of procedures have been developed for parameter estimation and inference in linear regression. These methods differ in computational simplicity of algorithms, presence of a closed-form solution, robustness with respect to heavy-tailed distributions, and theoretical assumptions needed to validate desirable statistical properties such as consistency and asymptotic efficiency. Some of the more common estimation techniques for linear regression are summarized below.

Linear regression -- Part 8

$$Y_{ij} = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_p X_{ip} + \varepsilon_{ij}$$

Linear regression -- Part 9

Epidemiology Early evidence relating tobacco smoking to mortality and morbidity came from observational studies employing regression analysis. In order to reduce spurious correlations when analyzing observational data, researchers usually include several variables in their regression models in addition to the variable of primary interest. For example, in a regression model in which cigarette smoking is the independent variable of primary interest and the dependent variable is lifespan measured in years, researchers might include education and income as additional independent variables, to ensure that any

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observed effect of smoking on lifespan is not due to those other socio-economic factors. However, it is never possible to include all possible confounding variables in an empirical analysis. For example, a hypothetical gene might increase mortality and also cause people to smoke more. For this reason, randomized controlled trials are often able to generate more compelling evidence of causal relationships than can be obtained using regression analyses of observational data. When controlled experiments are not feasible, variants of regression analysis such as instrumental variables regression may be used to attempt to estimate causal relationships from observational data.

Linear regression -- Part 10

which has an interpretation as the expected change in y' when all x_j' in the strongly correlated group increase by $(1/q)$ th of a unit at the same time with variables outside the group held constant. With strong positive correlations and in standardized units, variables in the group are approximately equal, so they are likely to increase at the same time and in similar amount. Thus, the average group effect ξ_A is a meaningful effect. It can be accurately estimated by its minimum-variance unbiased linear estimator $\hat{\xi}_A = 1/q (\hat{\beta}_1' + \hat{\beta}_2' + \dots + \hat{\beta}_q')$, even when individually none of the β_j' can be accurately estimated by $\hat{\beta}_j'$. Not all group effects are meaningful or can be accurately estimated. For example, β_1' is a special group effect with weights $w_1 = 1$ and $w_j = 0$ for $j \neq 1$, but it cannot be accurately estimated by $\hat{\beta}_1'$. It is also not a meaningful effect. In general, for a group of q strongly correlated predictor variables in an APC arrangement in the standardized model, group effects whose weight vectors $\mathbf{w}$ are at or near the centre of the simplex $\sum_{j=1}^q w_j = 1$ ($w_j \geq 0$) are meaningful and can be accurately estimated by their minimum-variance unbiased linear estimators. Effects with weight vectors far away from the centre are not meaningful as such weight vectors represent simultaneous changes of the variables that violate the strong positive correlations of the standardized variables in an APC arrangement. As such, they are not probable. These effects also cannot be accurately estimated. Applications of the group effects include (1) estimation and inference for meaningful group effects on the response variable, (2) testing for "group significance" of the q

variables via testing $H_0: \xi_A = 0$ versus $H_1: \xi_A \neq 0$, and (3) characterizing the region of the predictor variable space over which predictions by the least squares estimated model are accurate. A group effect of the original variables $\{x_1, x_2, \dots, x_q\}$ can be expressed as a constant times a group effect of the standardized variables $\{x_1', x_2', \dots, x_q'\}$. The former is meaningful when the latter is. Thus meaningful group effects of the original variables can be found through meaningful group effects of the standardized variables.